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12. The half-life of a radioactive isotope is 5 years. If you start with a 100 g sample, how much of the original isotope will remain after 15 years?

    A. 50 g

    B. 10 g

    **C. 12.5 g**

    D. 25 g

13. The half-life of a nuclide A is 6.93 days. How many days after a large sample of A has been isolated will it take for 90% of it to decay? ($\ln 2 = 0.693$ and $\ln 5 = 1.607$)

    **A. 23.03 days**

    B. 6.93 days

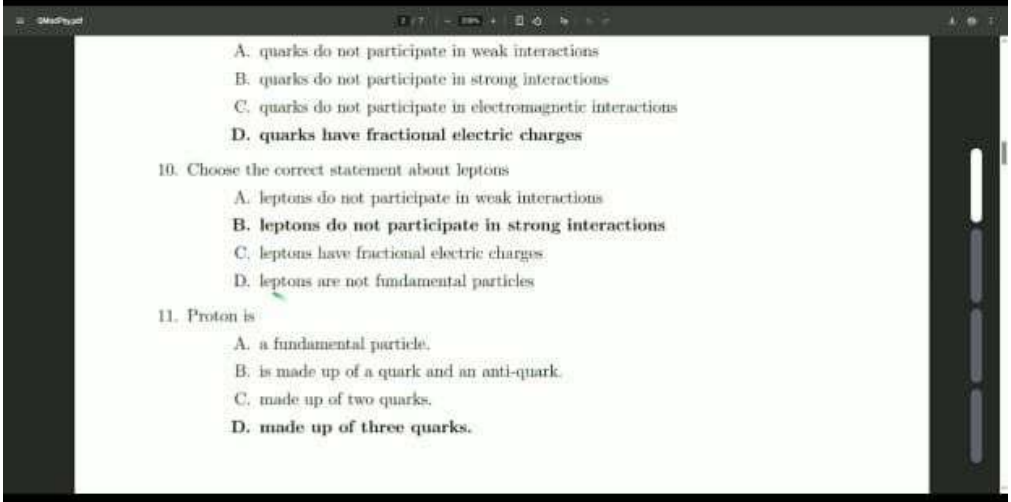
    C. 13.86 days

    D. 20.79 days

14. We express $S$ as a function of $E$, $V$, $N$, i.e, $S = S(E,V,N)$. We know $E \propto N$, $V \propto N$, $S \propto N$. We have $\mu = \left(\frac{\partial S}{\partial N}\right)_{E,V}$, i.e, $\mu$ is the partial derivative with respect

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A. quarks do not participate in weak interactions
B. quarks do not participate in strong interactions
C. quarks do not participate in electromagnetic interactions
**D. quarks have fractional electric charges**

10. Choose the correct statement about leptons
A. leptons do not participate in weak interactions
**B. leptons do not participate in strong interactions**
C. leptons have fractional electric charges
D. leptons are not fundamental particles

11. Proton is
A. a fundamental particle.
B. is made up of a quark and an anti-quark.
C. made up of two quarks.
**D. made up of three quarks.**

Saurabh
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D. Nuclei with even numbers of A

8. Which particle carries away the unobserved energy and momentum during the process of radioactive decay? A. neutron **B. neutrino** C. $\alpha$ particle D. $\beta$ particle

9. Choose the correct statement about quarks
   A. quarks do not participate in weak interactions
   B. quarks do not participate in strong interactions
   C. quarks do not participate in electromagnetic interactions
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   A. a fundamental particle.

Saurabh
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tering angle.

B. As the impact parameter of $\alpha$ increases, the scattering angle decreases.

C. The scattering cross-section is inversely proportional to the square of the kinetic energy of the $\alpha$ particle.

D. The scattering cross-section is proportional to the square of the atomic number of the foil-element.

6. The amount of scattering by a given foil is approximately proportional to the (velocity)$^a$ of the incident $\alpha$ particles. where 'a' is A. 2 B. 4 C. −2 **D. −4**

7. What are isobars?

**A. Nuclei with the same A, but different Z.**

B. Nuclei with the same number of neutrons.

C. Nuclei with same Z, but different A

D. Nuclei with even numbers of A

8. Which particle carries away the unobserved energy and momentum during the process of radioactive decay? A. neutron **B. neutrino** C. $\alpha$ particle D. $\beta$ particle

9. Choose the correct statement about quarks



Saurabh

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C. A majority of $\alpha$ particles are found to be back-scattered

D. $\alpha$ particles are absorbed before reaching the fluorescent screen detector.

5. Choose the **in**correct statement about the Rutherford's scattering experiment:

**A. The scattering cross-section varies as $\sin^4\theta/2$, where $\theta$ is the scattering angle.**

B. As the impact parameter of $\alpha$ increases, the scattering angle decreases.

C. The scattering cross-section is inversely proportional to the square of the kinetic energy of the $\alpha$ particle.

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1. In J. J. Thomson's experiment for measuring an electron's charge-to-mass (e/m) ratio, he first adjusted perpendicular B and E fields so that the electrons in the cathode ray tube were undeflected at the screen. The initial velocity of the electron will be equal to:
   A. e.B  B. e.E  **C. E/B**  D. B/E

2. How many 5 MeV electrons are required to deposit a total energy of 1J
   A. $5 \times 10^6$  **B.** $1.25 \times 10^{12}$  C. $8 \times 10^{13}$  D. $1.2 \times 10^{19}$

3. Find the energy of the electron at which its de-Broglie wavelength is comparable to the diameter of the $^{125}Te_{52}$ nuclei. (Use $hc = 1240$ MeV.fm)
   A. 124 MeV  B. 248 MeV  **C. 103.3 MeV**  D. 206.7 MeV

4. Choose the correct statement about the Rutherford's scattering experiment:
   **A. A majority of $\alpha$ particles are scattered at small angles.**
   B. A majority of $\alpha$ particles are found to be scattered at large angles (greater than 90°).



Saurabh

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- All the scribbling/rough work should be done in the provided rough sheets. No writing on the question paper is allowed.
- Non-programmable calculators allowed. No exchange of calculator allowed.
- For each correct answer 2 marks will be awarded.
- For each incorrect answer 1 marks will be deducted.

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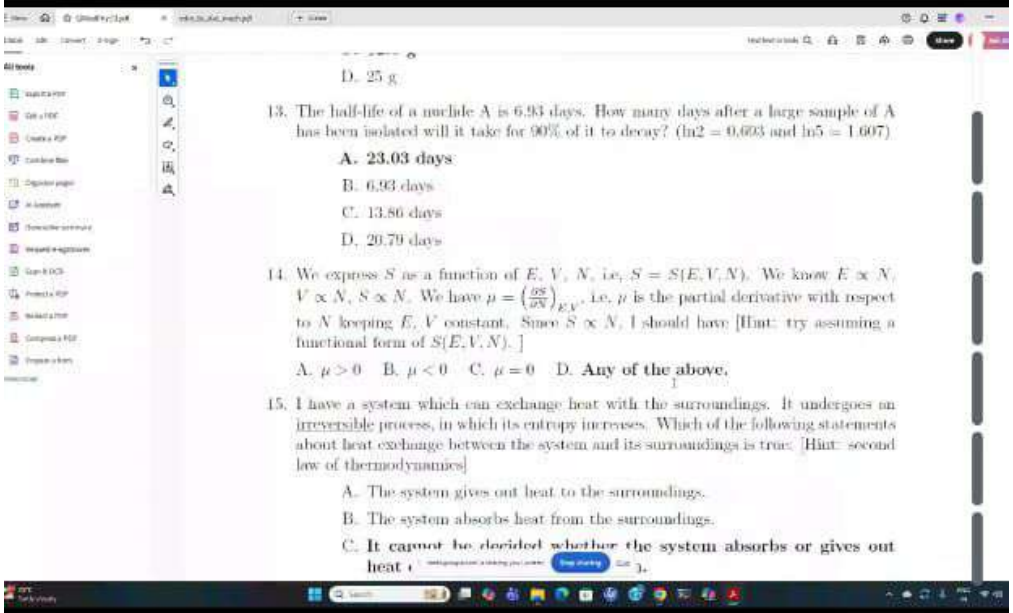
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Archak
Saurabh is presenting, annotating

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D. 25 g

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A. $\mu > 0$ B. $\mu < 0$ C. $\mu = 0$ D. **Any of the above.**

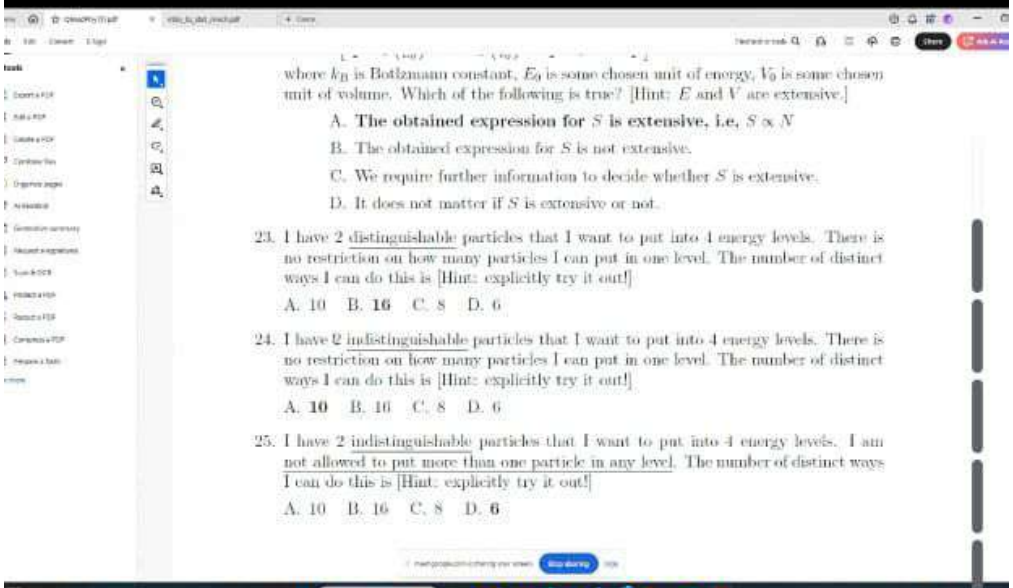
15. I have a system which can exchange heat with the surroundings. It undergoes an irreversible process, in which its entropy increases. Which of the following statements about heat exchange between the system and its surroundings is true. [Hint: second law of thermodynamics]

A. The system gives out heat to the surroundings.

B. The system absorbs heat from the surroundings.

C. **It cannot be decided whether the system absorbs or gives out heat (**

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where $k_B$ is Botlzmann constant, $E_0$ is some chosen unit of energy, $V_0$ is some chosen unit of volume. Which of the following is true? [Hint: $E$ and $V$ are extensive.]

A. **The obtained expression for $S$ is extensive, i.e. $S \propto N$**

B. The obtained expression for $S$ is not extensive.

C. We require further information to decide whether $S$ is extensive.

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23. I have 2 distinguishable particles that I want to put into 4 energy levels. There is no restriction on how many particles I can put in one level. The number of distinct ways I can do this is [Hint: explicitly try it out!]

A. 10 B. **16** C. 8 D. 6

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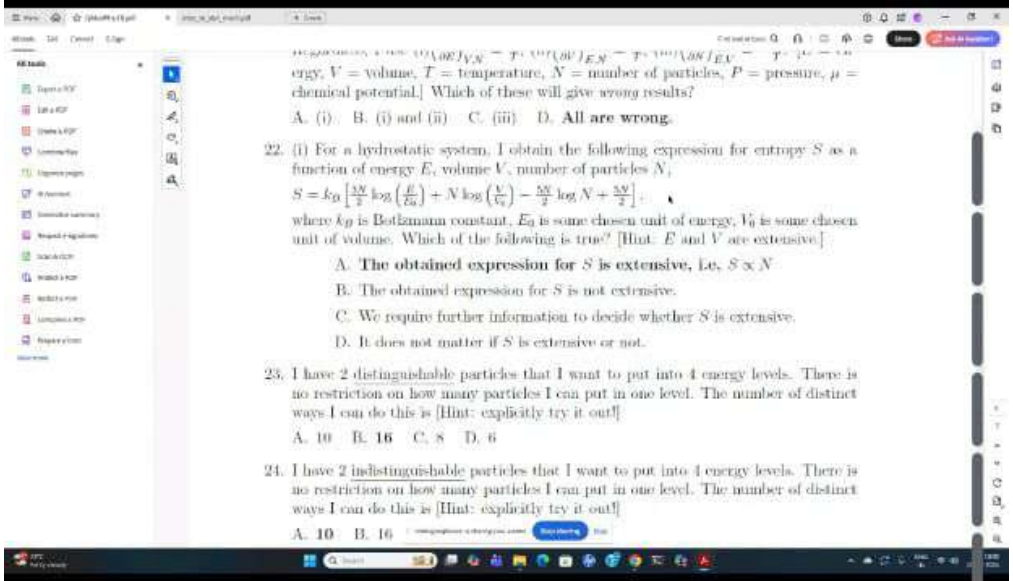
A. 10 B. 16 C. 8 D. **6**






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ergy, $V$ = volume, $T$ = temperature, $N$ = number of particles, $P$ = pressure, $\mu$ = chemical potential.] Which of these will give wrong results?

A. (i) B. (i) and (ii) C. (iii) D. **All are wrong.**

22. (i) For a hydrostatic system, I obtain the following expression for entropy $S$ as a function of energy $E$, volume $V$, number of particles $N$,

$$S = k_B \left[ \frac{5N}{2} \log \left( \frac{E}{E_0} \right) + N \log \left( \frac{V}{V_0} \right) - \frac{5N}{2} \log N + \frac{5N}{2} \right],$$

where $k_B$ is Boltzmann constant, $E_0$ is some chosen unit of energy, $V_0$ is some chosen unit of volume. Which of the following is true? [Hint: $E$ and $V$ are extensive.]

A. **The obtained expression for $S$ is extensive,** i.e. $S \propto N$

B. The obtained expression for $S$ is not extensive.

C. We require further information to decide whether $S$ is extensive.

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A. **10** B. 16

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$\mu$ = chemical potential.] Which of these will give *wrong* results?

A. (i) B. (i) and (ii) C. **(iii)** D. All are wrong.

21. To describe a system with 4 particles, I obtain the number of microstates $\Omega$ in equilibrium, assuming particles to be distinguishable. I calculate entropy using $S = k_B \log \Omega$, *without making use of Gibbs' idea*, and find $S$ to be *not extensive*.



Regardless, I use (i) $\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}$, (ii) $\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T}$, (iii) $\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}$. [$E$ = energy, $V$ = volume, $T$ = temperature, $N$ = number of particles, $P$ = pressure, $\mu$ = chemical potential.] Which of these will give *wrong* results?

A. (i) B. (i) and (ii) C. (iii) D. **All are wrong.**

22. (i) For a hydrostatic system, I obtain the following expression for entropy $S$ as a function of energy $E$, volume $V$, number of particles $N$,

$S = k_B \left[\frac{3N}{2} \log\left( \right) \quad \frac{5N}{2}\right]$,

where $k_B$ is Boltzmann constant. As $N$ varies either unit of energy, $V$ is cross-hatched

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A. $>0$ B. $<0$ C. $=0$ D. Any of the above.

19. Condisder gas in a box. The walls of the box are thermally conducting, i.e., *heat exchange is possible* between the gas in the box and the surrounding environment. The volume of the box is suddenly changed from initial value $V_i$ to final value $V_f$. Then the gas relaxes to a new equilibrium state. Let $\Omega$ be the number of microstates in final equilibrium state and $\Omega_{non-eq}$ be the number of microstates in the initial far-from-equilibrium state just after changing volume to $V_f$. $\Omega - \Omega_{non-eq}$ is [Hint: second law of thermodynamics, Boltzmann's law of Stat. Mech.]

A. $>0$ B. $<0$ C. $=0$ D. **Any of the above.**

20. To describe a hydrostatic system with 1 mole particles, I obtain the number of microstates $\Omega$ in equilibrium, assuming particles to be distinguishable. I calculat entropy using $S = k_B \log \Omega$, *without making use of Gibbs' idea*, and find $S$ to be *not extensive*. Regardless, I use (i) $\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}$, (ii) $\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T}$, (iii) $\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}$. [$E$ = energy, $V$ = volume, $T$ = temperature, $N$ = number of particles, $P$ = pressure, $\mu$ = chemical potential.] Which of these will give *wrong* results?

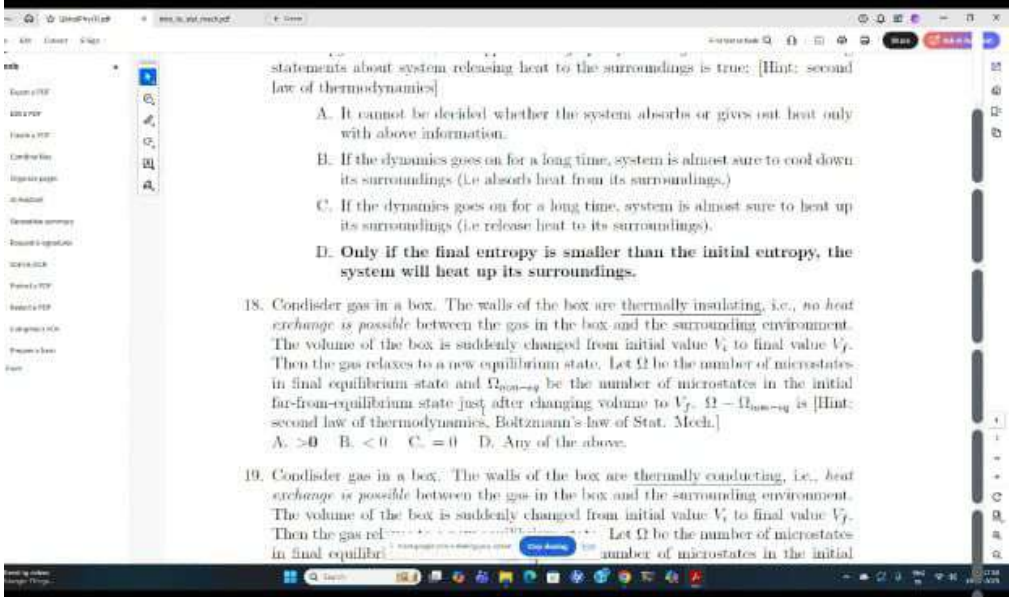
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Archak
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ceq-newj-wun



statements about system releasing heat to the surroundings is true: [Hint: second law of thermodynamics]

A. It cannot be decided whether the system absorbs or gives out heat only with above information.

B. If the dynamics goes on for a long time, system is almost sure to cool down its surroundings (i.e absorb heat from its surroundings.)

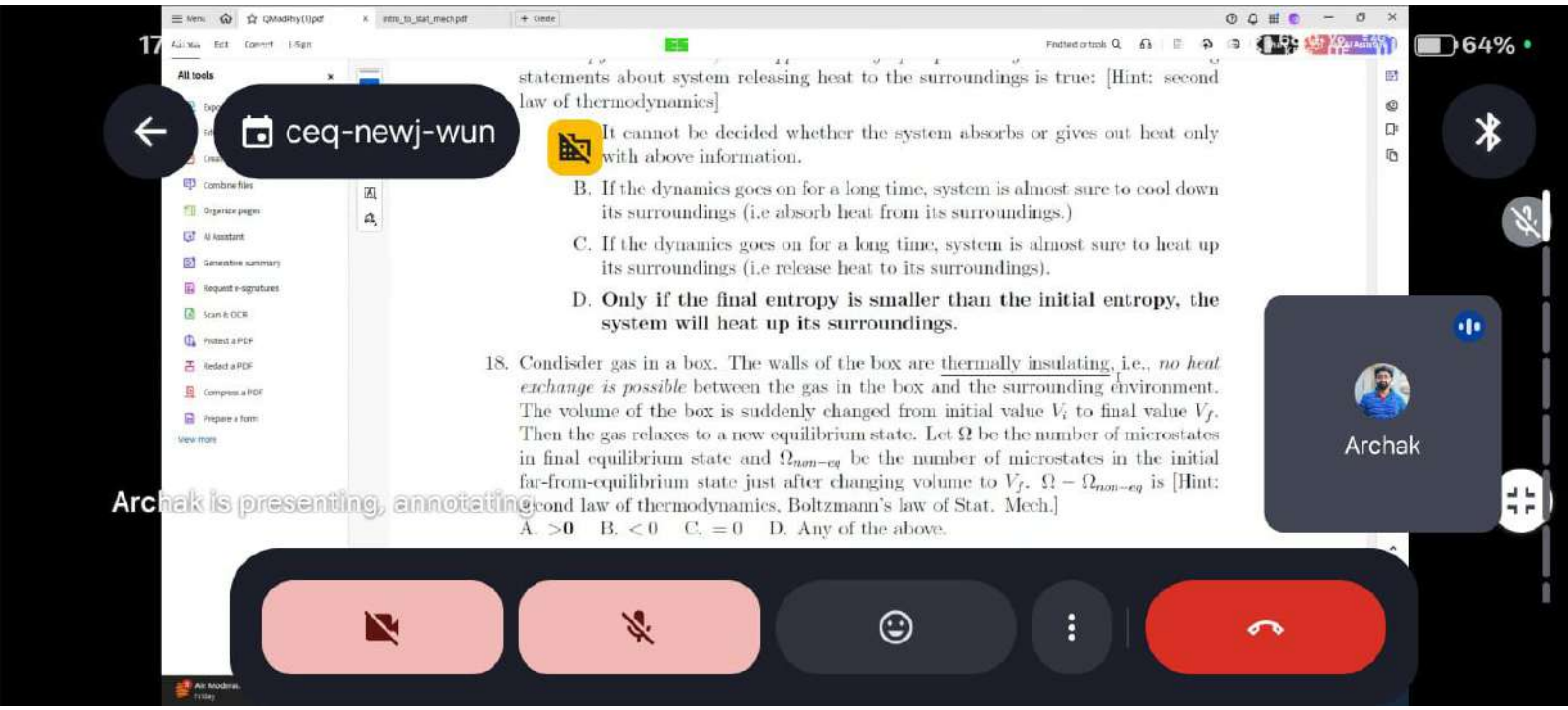
C. If the dynamics goes on for a long time, system is almost sure to heat up its surroundings (i.e release heat to its surroundings).

D. **Only if the final entropy is smaller than the initial entropy, the system will heat up its surroundings.**

18. Condisder gas in a box. The walls of the box are thermally insulating, i.e., *no heat exchange is possible* between the gas in the box and the surrounding environment. The volume of the box is suddenly changed from initial value $V_i$ to final value $V_f$. Then the gas relaxes to a new equilibrium state. Let $\Omega$ be the number of microstates in final equilibrium state and $\Omega_{non-eq}$ be the number of microstates in the initial far-from-equilibrium state just after changing volume to $V_f$. $\Omega - \Omega_{non-eq}$ is [Hint: second law of thermodynamics, Boltzmann's law of Stat. Mech.]
A. $>0$ B. $<0$ C. $=0$ D. Any of the above.

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A. $>0$ B. $<0$ C. $=0$ D. Any of the above.
Archak
Archak is presenting, annotating

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ceq-newj-wun

C. **It cannot be decided whether the system absorbs or gives out heat only with above information.**

D. There is no overall exchange of heat during the process.

16. I have a system which can exchange heat with the surroundings. It undergoes irreversible dynamics, which can be broken into small steps, each step is also irreversible. *During small steps of the dynamics, sometimes its entropy can decrease, sometimes its entropy can increase, with approximately equal probability.* Which of the following statements about system releasing heat to the surroundings is true: [Hint: second law of thermodynamics]

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C. Only if the final entropy of system is smaller than the initial entropy, the system will heat up its surroundings.



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52 others




D. It cannot be decided whether the system absorbs or gives out heat only with above information.

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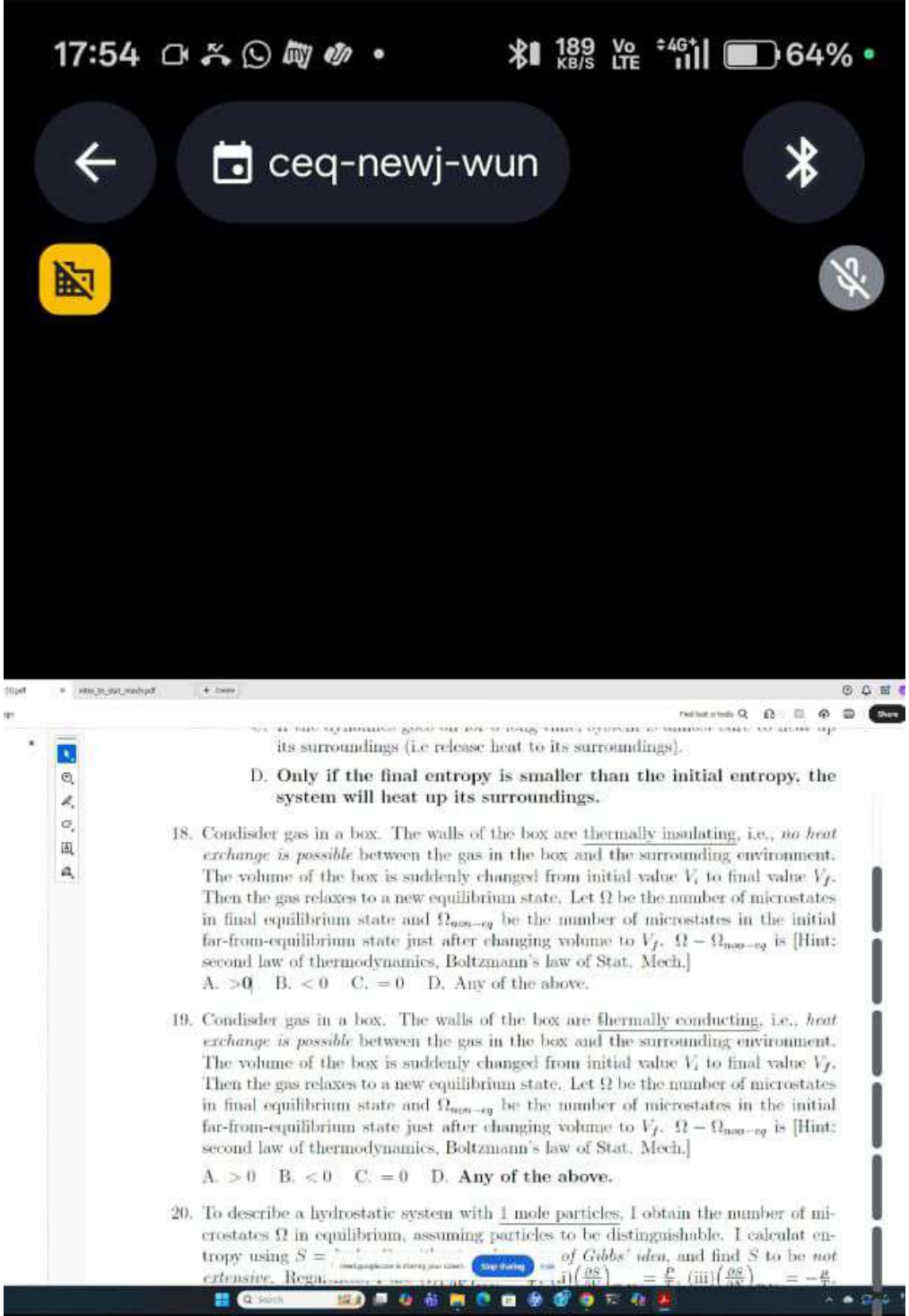
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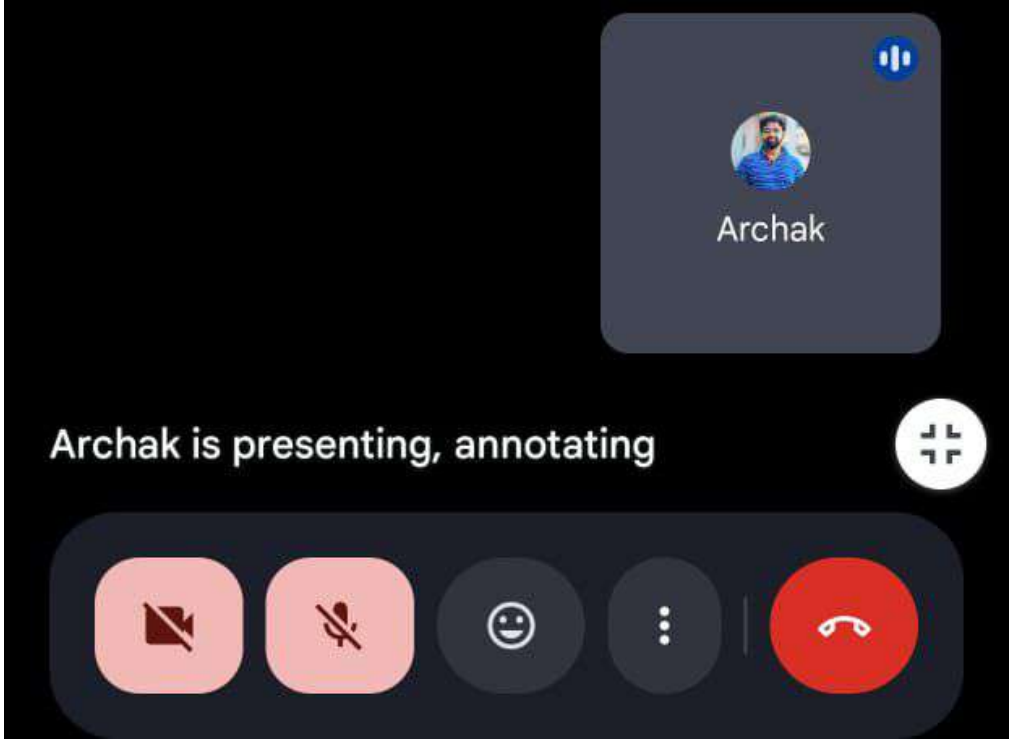

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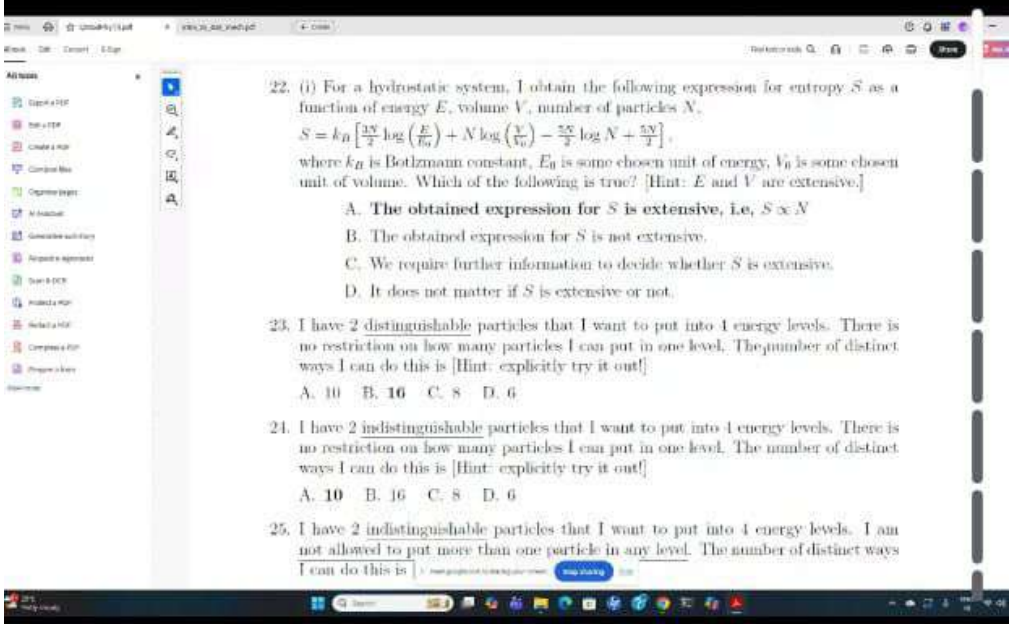
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Archak
Arepalli Neeraj Reddy joined

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4. Choose the correct statement about the Rutherford's scattering experiment:

   **A. A majority of $\alpha$ particles are scattered at small angles.**

   B. A majority of $\alpha$ particles are found to be scattered at large angles (greater than 90°).

   C. A majority of $\alpha$ particles are found to be back-scattered

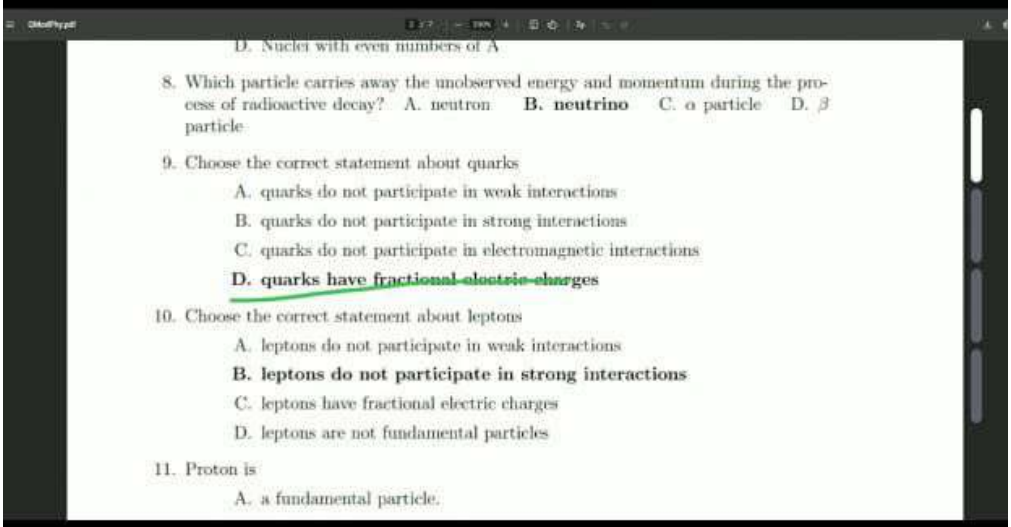
   D. $\alpha$ particles are absorbed before reaching the fluorescent screen detector.

5. Choose the **in**correct statement about the Rutherford's scattering experiment:

   **A. The scattering cross-section varies as $\sin^4\theta/2$, where $\theta$ is the scattering angle.**


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D. Nuclei with even numbers of A

8. Which particle carries away the unobserved energy and momentum during the process of radioactive decay? A. neutron **B. neutrino** C. $\alpha$ particle D. $\beta$ particle

9. Choose the correct statement about quarks
   A. quarks do not participate in weak interactions
   B. quarks do not participate in strong interactions
   C. quarks do not participate in electromagnetic interactions
   **D. quarks have fractional electric charges**

10. Choose the correct statement about leptons
   A. leptons do not participate in weak interactions
   **B. leptons do not participate in strong interactions**
   C. leptons have fractional electric charges
   D. leptons are not fundamental particles

11. Proton is
   A. a fundamental particle.




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